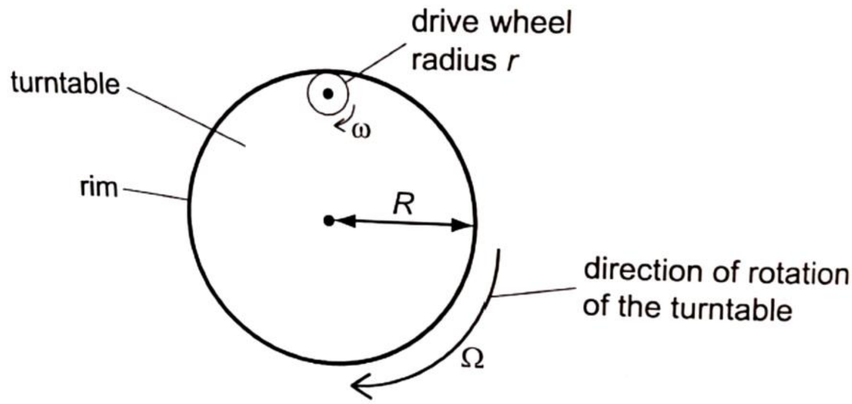
		($v = u + at$), hence, the power = Fv is increasing at a constant rate.

7 A turntable has radius R. It is driven by a rubber drive wheel of radius r in contact with the inside of the rim of the turntable, as shown below.



plan view

The turntable rotates with angular velocity Ω and the linear speed of a point on its rim is V. The drive wheel rotates with angular velocity ω and the linear speed of a point on its rim is v.

Which pair of equations show the relationship between the angular velocities and the linear speeds of the turntable and the wheel?

	angular velocities	linear speeds
A	$\Omega = \omega$	$V = v$
B	$\Omega = \omega$	$V = (\frac{r}{R})\, v$
C	$\Omega = (\frac{r}{R})\, \omega$	$V = v$
D	$\Omega = (\frac{R}{r})\, \omega$	$V = (\frac{r}{R})\, v$

7	**C**	Since no change in linear displacement, the 2 linear velocities are the same $R\,\Omega = r\ \omega$

[Turn over

8 The graph below shows the variation with distance x of the displacement y of a transverse